

ProjectDesignPhase-II

SolutionRequirements(Functional&Non-functional)

Metric Details	
Date	08 July 2026
Team ID	K. Bhargav
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Dashboard Visualization	Display interactive Tableau dashboard with key metricsDisplay KPI summary cards for total observations, disease prevalence, and average BMI
FR-2	Demographic Risk Analysis	Analyze demographic-wise clinical risk demandCompare age and biological sex vulnerabilities across different cohorts
FR-3	Behavioral Analysis	Display disease distribution by smoking habitsShow risk contribution by alcohol consumption and concurrent usage
FR-4	Comorbidity Mapping	Analyze cardiovascular events by varying diabetic severityCompare critical stroke occurrences across different metabolic segments
FR-5	Secondary Ailment Profiling	Display overlap trends of asthma, kidney disease, and skin cancerCompare secondary ailment performance across racial demographics
FR-6	Metabolic Risk Analysis	Analyze severe BMI threshold volumesAnalyze body mass vectors mapped across age categories
FR-7	Interactive Filters	Apply filters by demographic, clinical status, and lifestyle factorsEnable drill-down into individual visualizations

FR No.	Functional Requirement	Sub Requirement (Story / Sub-Task) (Epic)
FR-8	Tableau Story	Display a guided analytical story of clinical risk trends Present key medical insights in a narrative format
FR-9	Flask Website	Embed the Tableau dashboard within the Flask website Embed the Tableau story within the Flask website
FR-10	Business Reporting	Present summarized clinical insights for hospital stakeholders Support data-driven decision making for diagnostic specialists and clinical managers

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to use, with clear navigation and intuitive interactive visualizations for clinical users.
NFR-2	Security	The application should securely present analytics through the Flask website without exposing raw or sensitive medical data.
NFR-3	Reliability	The dashboard should consistently produce accurate and dependable analytical results from the underlying diagnostic data.
NFR-4	Performance	Dashboard pages should load efficiently, and visualizations and filters should respond smoothly to user interaction.
NFR-5	Availability	The Tableau dashboard and Flask website should remain accessible to users whenever clinical insights are needed.
NFR-6	Scalability	The solution should be able to accommodate larger EHR datasets and additional dashboards as the project grows in future.